CHANG, HUNG-CHUN

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Education

National Taiwan University **Feb. 2025–present**
M.S. Department of Psychology GPA: 3.87/4.3
Master's Thesis Research (in progress): Validity of LLMs as Simulated Students

National Taiwan University **Sep. 2019–Jan. 2025**
B.S. in Interdisciplinary Studies of Human-Computer Interaction, University Interdisciplinary Bachelor's Program
B.S. in Agriculture, Department of Forestry and Resource Conservation; Minor in Psychology
Thesis: Design and Evaluation of a Conversational Robot for the Elderly Based on Social Penetration Theory

Research Experience

Social Lab Human-Robot Interaction group *Research Assistant* **Nov. 2022–present**
↪ Prof. Hsiu-Ping Yueh, Weijane Lin *Department of Psychology, National Taiwan University*

- **Held** 2 workshops on multi-modal data analysis and voice-based social robot, 4 workshops on social robotic programming.
- **Instructed** 3 undergraduate students on their projects.

Learning and Educational Technologies Research Unit *Research Intern* **Jun. 2026–Aug. 2026**
↪ Prof. Hiroaki Ogata, Chia-Yu Hsu *Academic Center for Computing and Media Studies, Kyoto University*

- **Conducted** an international collaboration project on simulated students, accepted by ReLEAF@ICCE2026.

Social Robotics Classes *Teaching Assistant* **Feb. 2025–present**
↪ Prof. Hsiu-Ping Yueh *Department of Psychology, National Taiwan University*

- **Led** TA classes over **18** hours across **4** semesters, including literature review, user research, prototype and program skills for designing social robotic systems.

Publications

H.-C. Chang, C.-Y. Hsu, W. Lin, H. Ogata, and H.-P. Yueh. Does an ability-conditioned LLM student simulator hold up as a construct? In *Proceedings of the 1st Workshop on Trustworthy Educational Data Sharing and Secondary Use: The ReLEAF Data Challenge, 34th International Conference on Computers in Education (ICCE 2026)*, 2026. Accepted; to appear.

H.-P. Yueh, **H.-C. Chang**, J.-H. Lo, S.-H. Wang, H.-Y. Chang, P.-S. Yang, and H.-P. Huang. What makes a robot feel affective? verbal cue types and the role of agreeableness in older adults' perception of robot personality. *International Journal of Human–Computer Interaction*, pages 1–22, 2026a. doi: 10.1080/10447318.2026.2722790.

H.-P. Yueh, **H.-C. Chang**, G.-Y. Wang, P.-L. Hung, and W. Lin. Adaptive empathy evaluation: an intelligent social robot tutor with multimodal assessment. *Enterprise Information Systems*, 2026b. doi: 10.1080/17517575.2026.2683825.

H.-P. Yueh, P.-L. Hung, G.-Y. Wang, **H.-C. Chang**, and W. Lin. Empathic AI-powered cognitive robots as tutors: Assessing their impact on learning performance in accelerated learning contexts. *Educational Psychology*, 2026c. Under revision.

H.-C. Chang and H.-P. Yueh. Assessment and evaluation of self-disclosure, application of behavior construct and semantic analysis. Paper presented at the 16th International Conference of Digital Archives and Digital Humanities (DADH 2025), National Taiwan University, Taipei, Taiwan, November 2025.

Research Projects

Master's Thesis Research (in progress) **Feb. 2026–present**
Validity of LLMs as Simulated Students. *Prof. Hsiu-Ping Yueh, Feng-Ming Tsao and Hiroaki Ogata*

- **Recast** the validity of an ability-conditioned LLM student simulator as a nomological network of four behavioral evidences: profile–behavior consistency, fidelity of error, boundary of competence, and stateful learning.
- **Tested** all four concurrently against real per-tier learner data from the PSLC DataShop Fractions 2012 corpus, using zero-shot prompting and a 2 × 2 fine-tuning ablation (reasoning × history).
- **Found** that fine-tuning with reasoning and history simulated students best, yet the competence paradox persisted and the learning mechanism was not reproduced; this first part was accepted at ReLEAF@ICCE 2026 (first author).

ECRobot, an ITS robot *Sole developer* **Sep. 2025–Dec. 2025**
Enterprise Information Systems, 2026; Educational Psychology, under revision

- **Designed** student simulation pipeline and data analysis, evaluation method and wrote the first draft (Enterprise Information Systems).
- **Instantiated** prospect theory as an empathetic feedback algorithm for the robot tutor (Educational Psychology).
- **Built** the entire tutoring system alone, including an iPad app (Swift) for handwriting input, a NAO robot (Qi Python) as voice user interface (VUI), and a FastAPI backend for organizing information and providing feedback.

Companion robot for older adults *Data Analysis & Front-end Development* **Mar. 2024–Dec. 2025**
International Journal of Human-Computer Interaction, 2026

- **Operationalized** discourse affectiveness into measurable linguistic indicators, and validated the manipulation with NLP-based checks.
- **Developed** the front-end app that presents the robot's responses.
- **Sustained** conversations averaging 15 minutes per session and 30 minutes per participant.

Social Robot Design Platform *Initiator & sole developer* **Feb. 2025–present**

- **Translated** social robotics system development workflows into a Social Robot Design Language, inspired by my experience of developing 5 social robotic systems.
- **Implemented** the design language as a web-based end-user programming tool.
- **Collaborated** with four groups of student researchers in a research-through-design process over 6 months.
- **Deployed** the system in the laboratory experiment for over a month.

TouchHear, an AR assistive learning system *Sole developer* **Sep. 2025–Dec. 2025**

- **Replaced** the experimenter-in-the-loop step of the original protocol — a researcher watched the participant's hand and cued audio manually — with real-time fingertip tracking on a depth camera.
- **Built** an authoring interface that turns any printed image into a touch-responsive surface: non-programmers draw regions, map audio to them, and export a printable template.
- **Designed** the printed template's ArUco markers to do double duty, carrying both depth-plane calibration and project identity, making the sheet self-describing.

Honors and Funding

International PhD/Master's Students Co-Advising Project *NT$250,000.* **Jun. 2026–Aug. 2026**

- Initiated the collaboration and secured the host placement independently; funded a three-month residency at Ogata's LET Lab, Kyoto University, co-advised by Prof. Hiroaki Ogata and Prof. Chia-Yu Hsu.

NTU College of Science Travel Grant and Scholarship *NT$15,000.* **Jun. 2026–Aug. 2026**

- Travel support from the College of Science, National Taiwan University.

Cathay Charity Foundation 2025 Excellence Scholarship *NT$160,000.* **Dec. 2025**

- Supported the companion robot for older adults.

Selected Course Work

Psychology & Methods	Robotics	Human-Computer Interaction
Psychoinformatics and Neuroinformatics	Introduction to Robotics	HCI Design
Neural and Behavioral Modeling	Robot Vision	HCI and User Research
Applied Linear Regression	Natural Language Processing	Gerontechnology Design Seminar
Experimental Design	Social Robots	Persuasive Social Robotics
Statistics in Psychology and Education	Social Robot Special Topics	Interactive Design

Skills

Programming Languages Python, Java, Swift, Matlab, LaTeX & TeX
Languages Mandarin (native), English (fluent), Japanese (JLPT N3)